Class 18: Pertussis Mini Project

Nathan Joseph (PID:A17668656)

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Background

Pertussis (a.k.a. whooping cough) is a common lung infection caused by the bacteria B. Pertussis.

This can affect all ages but is most severe for those under 1 year of age.

The CDC tracks the number of reported cases in the US:

We can "scrape" this data with the **datapasta** package.

```
cdc <- data.frame(
  year = c(
    1922L,
    1923L,
    1924L,
    1925L,
    1926L,
    1927L,
    1928L,
    1929L,
    1930L,
    1931L,
    1932L,
    1933L,
    1934L,
```

```
    1935L,
    1936L,
    1937L,
    1938L,
    1939L,
    1940L,
    1941L,
    1942L,
    1943L,
    1944L,
    1945L,
    1946L,
    1947L,
    1948L,
    1949L,
    1950L,
    1951L,
    1952L,
    1953L,
    1954L,
    1955L,
    1956L,
    1957L,
    1958L,
    1959L,
    1960L,
    1961L,
    1962L,
    1963L,
    1964L,
    1965L,
    1966L,
    1967L,
    1968L,
    1969L,
    1970L,
    1971L,
    1972L,
    1973L,
    1974L,
    1975L,
    1976L,
```

```
    1977L,
    1978L,
    1979L,
    1980L,
    1981L,
    1982L,
    1983L,
    1984L,
    1985L,
    1986L,
    1987L,
    1988L,
    1989L,
    1990L,
    1991L,
    1992L,
    1993L,
    1994L,
    1995L,
    1996L,
    1997L,
    1998L,
    1999L,
    2000L,
    2001L,
    2002L,
    2003L,
    2004L,
    2005L,
    2006L,
    2007L,
    2008L,
    2009L,
    2010L,
    2011L,
    2012L,
    2013L,
    2014L,
    2015L,
    2016L,
    2017L,
    2018L,
```

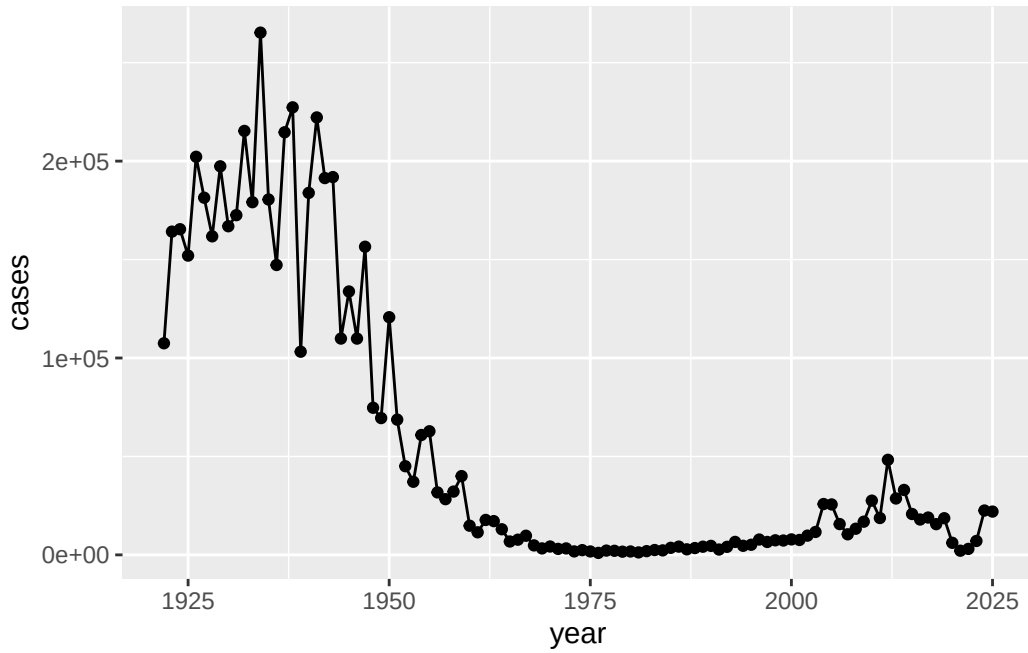
```
    2019L,
    2020L,
    2021L,
    2022L,
    2023L,
    2024L,
    2025L
  ),
  cases = c(
    107473,
    164191,
    165418,
    152003,
    202210,
    181411,
    161799,
    197371,
    166914,
    172559,
    215343,
    179135,
    265269,
    180518,
    147237,
    214652,
    227319,
    103188,
    183866,
    222202,
    191383,
    191890,
    109873,
    133792,
    109860,
    156517,
    74715,
    69479,
    120718,
    68687,
    45030,
    37129,
    60886,
```

```
    62786,
    31732,
    28295,
    32148,
    40005,
    14809,
    11468,
    17749,
    17135,
    13005,
    6799,
    7717,
    9718,
    4810,
    3285,
    4249,
    3036,
    3287,
    1759,
    2402,
    1738,
    1010,
    2177,
    2063,
    1623,
    1730,
    1248,
    1895,
    2463,
    2276,
    3589,
    4195,
    2823,
    3450,
    4157,
    4570,
    2719,
    4083,
    6586,
    4617,
    5137,
    7796,
```

```
    6564,
    7405,
    7298,
    7867,
    7580,
    9771,
    11647,
    25827,
    25616,
    15632,
    10454,
    13278,
    16858,
    27550,
    18719,
    48277,
    28639,
    32971,
    20762,
    17972,
    18975,
    15609,
    18617,
    6124,
    2116,
    3044,
    7063,
    22538,
    21996
  )
)
```

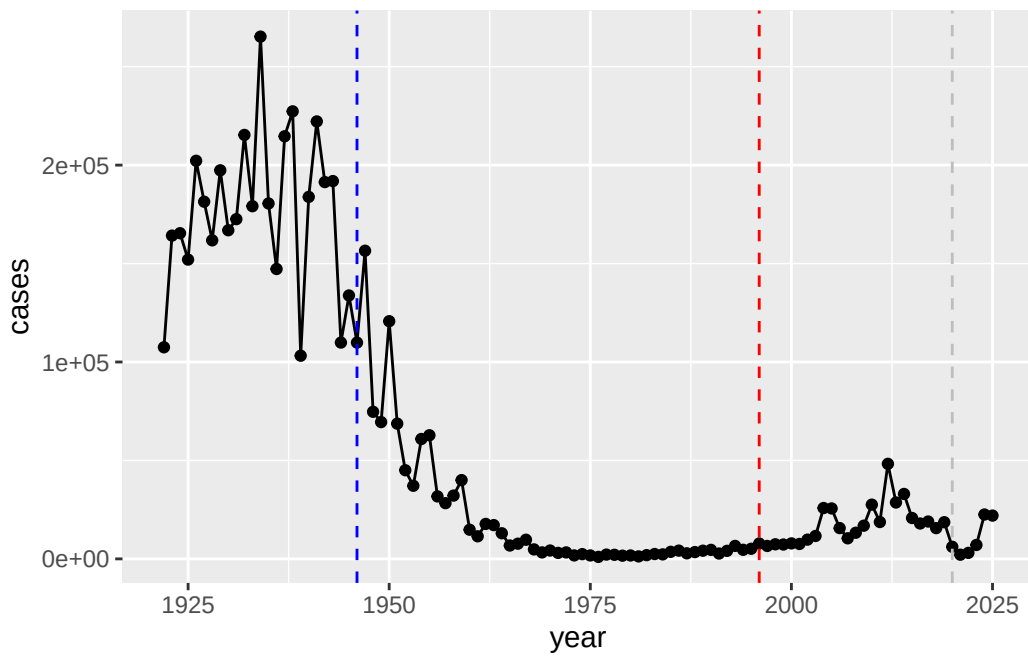
Q. Make a plot of `year` vs `cases`

```
library(ggplot2)
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line()
```



Q. Add some major milestones including the first wP vaccine roll-out (1946), the switch to the newer aP vaccine (of 1996), the COVID years (2020)?

```
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, col="blue", lty=2) +
  geom_vline(xintercept = 1996, col="red", lty=2) +
  geom_vline(xintercept = 2020, col="gray", lty=2)
```



There were high case numbers in the pre-1940s, then the introduction of the first wP vaccine in 1946 lowered the case numbers and then the case numbers got even lower with the introduction of the second aP vaccine in 1996. However in the mid-2000s, the case numbers increased due to circulating myths regarding vaccines and more people lacking the booster update shots but the case numbers lowered in 2020 due to quarantine and lockdown mandates which prevented transmission of the whooping cough.

Why is this vaccine-preventable disease on the upswing? To answer this question we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated individuals.

CMI-PB Project

Computational Models of Immunity - Pertussis Boost project aims to provide the scientific community with this very information.

They make their data available vis JSON format returning API. We can read this in R with the `read_json()` function from the **jsonlite** package:

```
library(jsonlite)

subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
```

```r
head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q. How many wP and aP individuals are in this `subject` table?

```r
sum(subject$infancy_vac == "wP")
```

```
[1] 85
```

```r
sum(subject$infancy_vac == "aP")
```

```
[1] 87
```

```r
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q. What is the biological sex breakdown?

```r
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q. In terms of race and gender, is this dataset representative of the US population?

```r
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

Let's read some more tables:

```r
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = T)

ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = T)
```

```r
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```r
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To analyze this data, we need to first "join" (merge/link) the different tables so we have all the data in one place not spread across different tables.

We can use the `*_join()` family of functions from **dplyr** to do this

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

```r
abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
```

```
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q. What antibody isotypes are measured for these patients?

```r
table(ab_titer$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Q. What antigens are reported?

```r
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA 
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318 
    PD1     PRN      PT     PTM   Total      TT 
   1970    6712    6712    1970     788    6318 
```

Let's focus on the IgG isotype and make a plot of MFI_normalized for all antigens

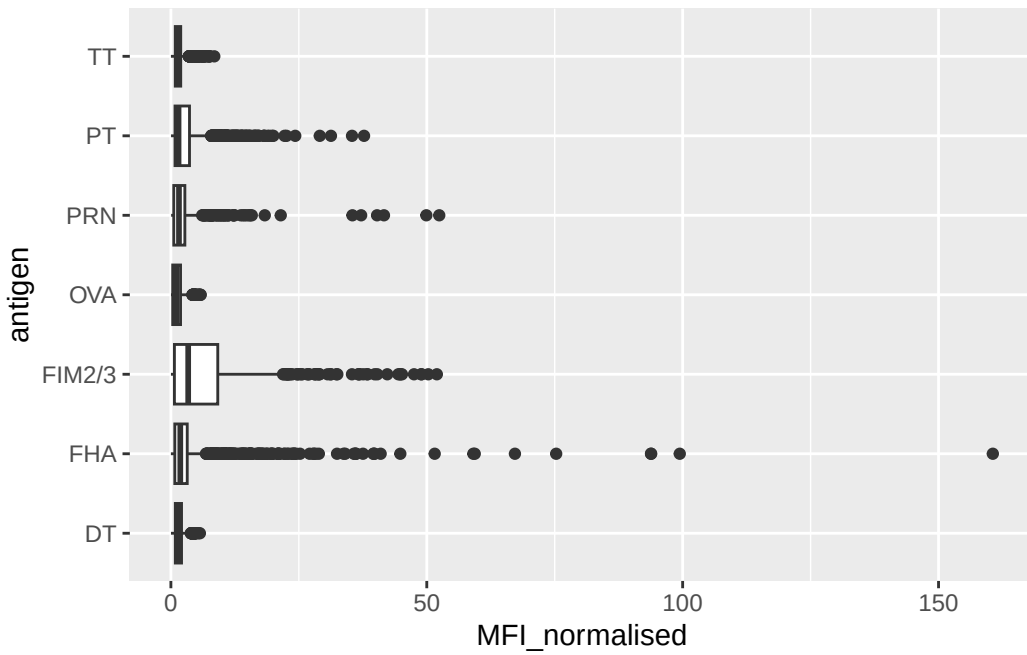
```r
igg <- abdata |>
  filter(isotype == "IgG")

head(igg)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
6 IU/ML                 4.679535          3          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
```

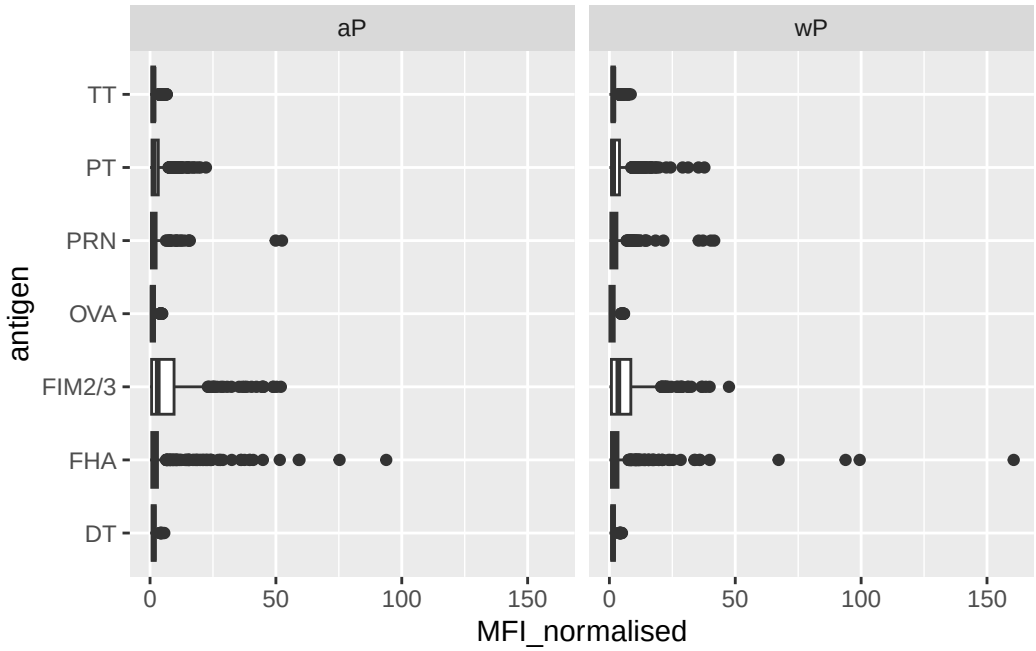
```
5                                -3                                0          Blood
6                                -3                                0          Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```

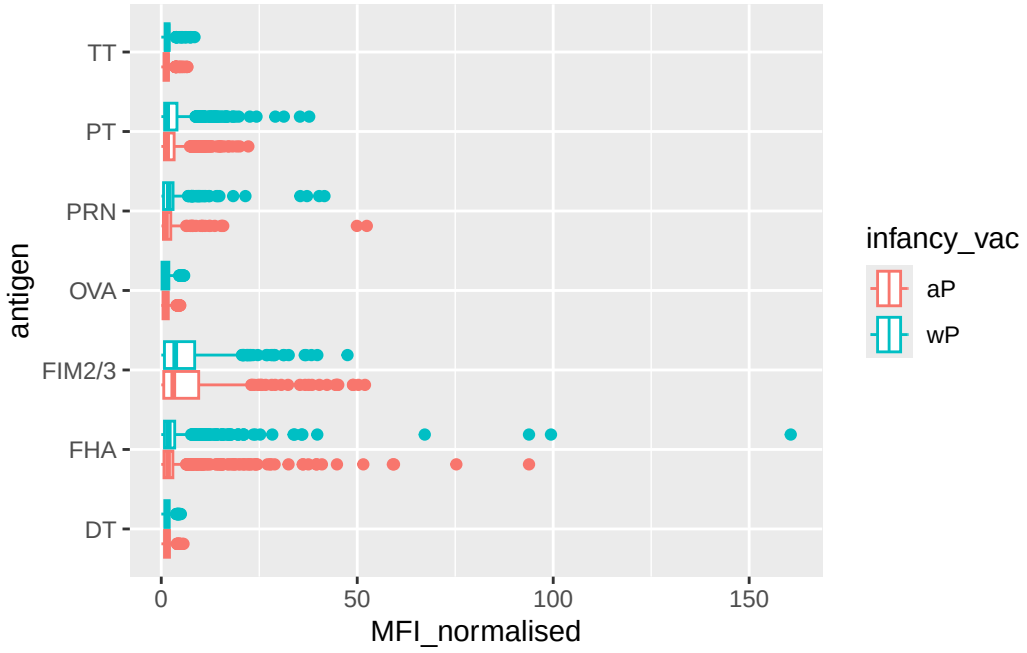


Q. Is there a difference for aP vs wP individuals with these values?

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```

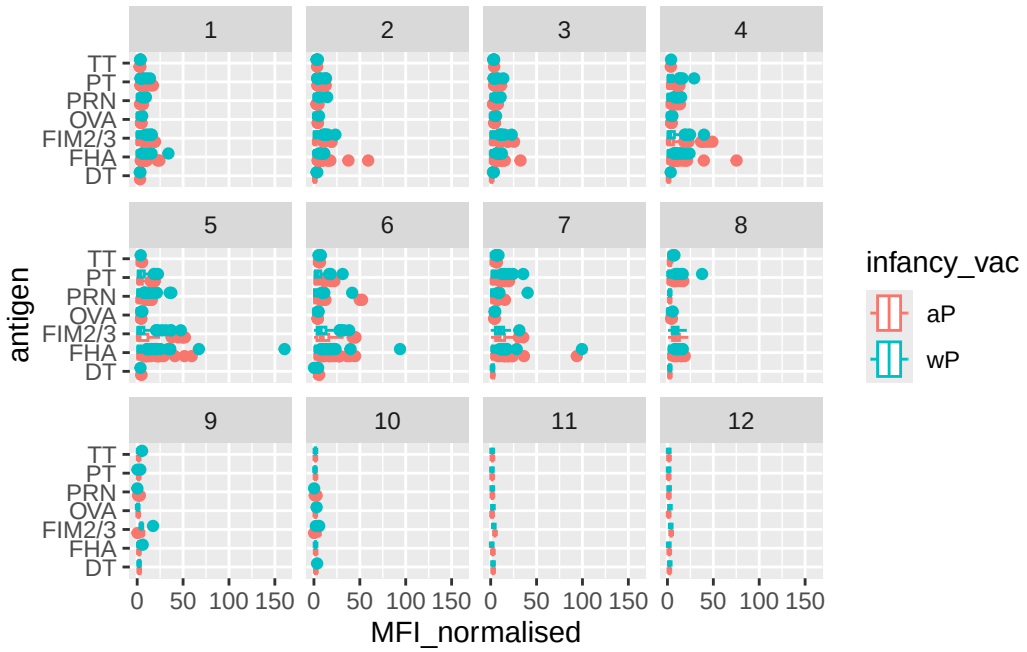



```
ggplot(igg) +
  aes(MFI_normalised, antigen, col = infancy_vac) +
  geom_boxplot()
```



Q. Is there a temporal response - i.e. do values increase or decrease over time?

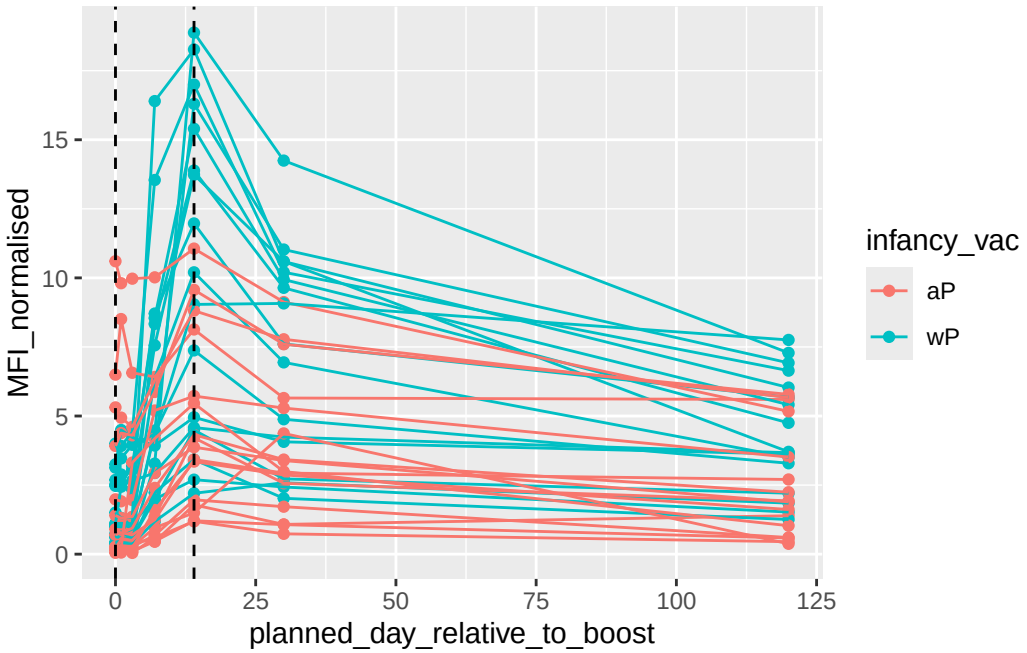
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col = infancy_vac) +
  geom_boxplot() +
  facet_wrap(~visit)
```



Focus on "PT" Pertussis Toxin antigen

```
pt.igg.21 <- igg |> filter(antigen == "PT",
              dataset == "2021_dataset")
```

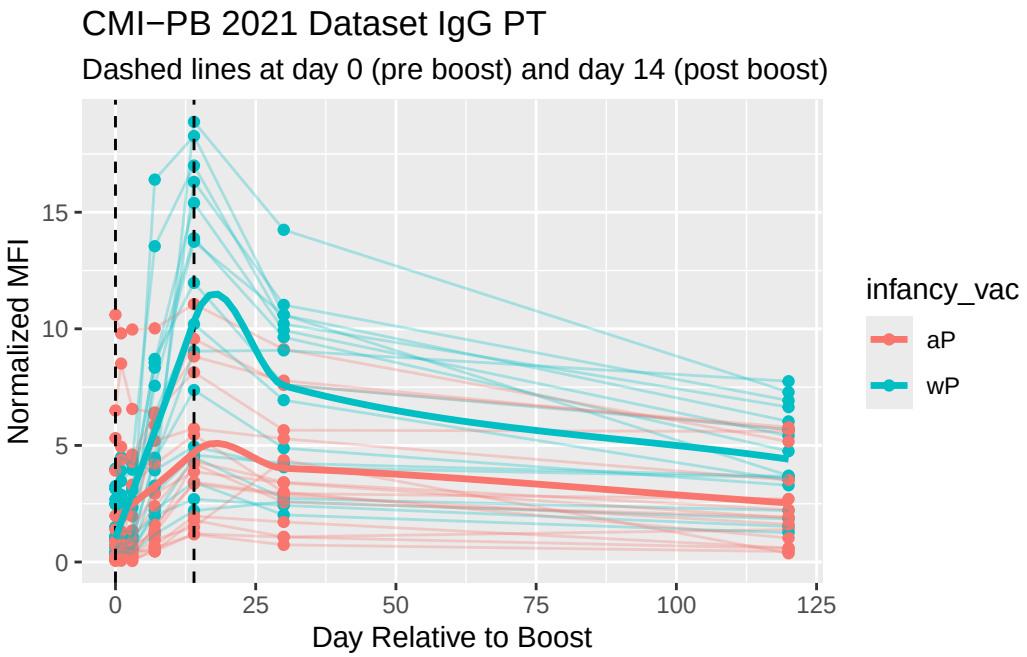
```
ggplot(pt.igg.21) +
  aes(planned_day_relative_to_boost,
      MFI_normalised,
      col = infancy_vac,
      group = subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 14, lty=2) +
  geom_vline(xintercept = 0, lty=2)
```




```
ggplot(pt.igg.21) +
  aes(planned_day_relative_to_boost,
      MFI_normalised,
      col = infancy_vac,
      group = subject_id) +
  geom_point() +
  geom_line(alpha = 0.3) +
  geom_vline(xintercept = 14, lty=2) +
  geom_vline(xintercept = 0, lty=2) +
  geom_smooth(aes(group = infancy_vac), se = FALSE, size = 1.2,
              span = 0.75, method.args = list(degree = 1)) +
  labs(
    title = "CMI-PB 2021 Dataset IgG PT",
    subtitle = "Dashed lines at day 0 (pre boost) and day 14 (post boost)",
    x = "Day Relative to Boost",
    y = "Normalized MFI"
  )
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.

`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```


CMI−PB 2021 Dataset IgG PT
Dashed lines at day 0 (pre boost) and day 14 (post boost)
Normalized MFI
15
10
5
0
0
25
50
75
100
125
Day Relative to Boost
infancy_vac
aP
wP